

"Can SAP HANA Replace Traditional RDBMS? A Critical Analysis of In-Memory Databases for Real-Time Enterprise Analytics"

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Abstract — The rapid growth of enterprise data has significantly increased the demand for faster and more integrated data processing systems. Traditional disk-based Relational Database Management Systems (RDBMS) remain reliable for transactional workloads, but they face limitations when organizations require real-time analytical insights from large-scale operational data. These limitations are not only caused by hardware latency, but also by the architectural separation between Online Transaction Processing (OLTP) and Online Analytical Processing (OLAP), which often requires delayed batch-based ETL processes.

This paper examines SAP HANA as an in-memory database platform that attempts to address these limitations through RAM-based processing, columnar storage, data compression, and the integration of transactional and analytical workloads within a single system. The study first compares the theoretical architectural differences between SAP HANA and traditional RDBMS, then evaluates the practical value of SAP HANA through the Walmart case study. The case demonstrates how in-memory computing can support real-time supply chain analytics, faster query execution, and more responsive decision-making. However, the transformation also involves challenges such as high infrastructure cost, migration complexity, and the need to determine which data should remain in memory.

The analysis argues that SAP HANA should not be viewed as a complete replacement for all traditional RDBMS environments. Instead, it is most suitable for enterprises that require real-time analytics and can justify the cost of in-memory infrastructure. Therefore, the value of SAP HANA depends on workload requirements, data volume, business urgency, and cost-performance trade-offs.

Keywords — SAP HANA, Traditional RDBMS, In-Memory Databases, Real-Time Analytics, OLTP/OLAP Integration, Database Architecture, Performance Trade-offs

I. INTRODUCTION

A. General Background and Motivation

The rapid expansion of enterprise data in the digital era has significantly increased the demand for systems capable of processing both transactional and analytical workloads efficiently (Zeier et al., 2011; Chen et al., 2014). Traditional Relational Database Management Systems (RDBMS) have long been the foundation of enterprise data management due to their reliability and consistency in handling Online Transaction Processing (OLTP) workloads (Plattner, 2009). However, as organizations increasingly rely on data-driven decision-making, the need for real-time analytics has exposed inherent limitations in conventional database architectures (Chaudhuri et al., 2011).

These limitations are not solely due to hardware constraints such as disk Input/Output (I/O) latency, but more importantly arise from the architectural design of traditional systems (Plattner, 2009). In particular, the strict separation between OLTP and Online Analytical Processing (OLAP) environments requires complex and time-consuming Extract, Transform, and Load (ETL) processes (Vassiliadis, 2009). As a result, organizations often depend on delayed, batch-processed data rather than real-time insights, which reduces their ability to respond quickly to dynamic business conditions (Zeier et al., 2011).

B. Problem Statement

To address these limitations, in-memory database systems such as SAP HANA have been introduced (Plattner, 2014). By storing data in main memory and utilizing columnar storage, SAP HANA enables the integration of transactional and analytical workloads within a single platform (Färber et al., 2012; Sikka et al., 2012). This architectural shift allows for significantly faster query execution and supports real-time analytics (Plattner, 2014). However, while SAP HANA demonstrates substantial performance advantages, it also introduces new challenges, including high infrastructure costs and scalability concerns associated with enterprise-grade memory (Lahiri et al., 2016).

Therefore, an important question arises: can in-memory database systems such as SAP HANA fully replace traditional RDBMS, or should they be considered as complementary solutions depending on specific enterprise requirements?

C. Objective and Contribution

This paper aims to provide a critical evaluation of SAP HANA in comparison with traditional RDBMS. First, it analyzes the fundamental architectural differences between disk-based and in-memory database systems, focusing on storage mechanisms, workload integration, and performance characteristics (Plattner, 2009; Färber et al., 2012). Second, it examines the practical applicability of SAP HANA through a real-world case study of Walmart, highlighting both the benefits and challenges encountered during its implementation (Smith & Kumar, 2018). Finally, the paper evaluates whether SAP HANA can serve as a complete replacement for traditional RDBMS or whether its adoption should be determined based on specific business needs, cost considerations, and data processing requirements.

II. CONTENT DISCUSSION

A. SAP HANA Architecture and Core Mechanisms

The performance improvements and real-time capabilities offered by SAP HANA are not incremental enhancements but are driven by a fundamentally different database architecture.

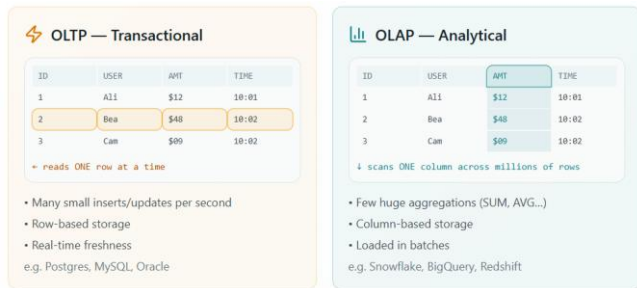


Figure 1: Comparison of Row-Based and Column-Based Storage

Figure 1 illustrates the fundamental difference between row-based and column-based storage, which represents one of the key architectural distinctions between traditional RDBMS and SAP HANA. In traditional RDBMS, data is commonly stored in a row-based format, where all attributes of a complete record are stored together. This structure is suitable for Online Transaction Processing (OLTP) workloads because the system can efficiently insert, update, delete, or retrieve a full transaction record (Plattner, 2009).

However, row-based storage is less efficient for analytical workloads because analytical queries usually require only selected columns across a large number of records. For example, a query calculating total sales or average transaction amount may only need the amount column, but a row-based system may still need to scan additional unnecessary data. This increases data access and slows down query performance when handling large datasets (Binnig et al., 2009; Färber et al., 2012).

In contrast, SAP HANA adopts a column-based storage model, where data is organized by attributes rather than complete records. This allows analytical queries to access only the relevant columns required for computation, instead of scanning entire rows. As a result, columnar storage reduces unnecessary data scanning, improves query performance, enables better data compression, and is particularly suitable for large-scale analytical workloads (Binnig et al., 2009; Färber et al., 2012).

This difference in storage organization explains why SAP HANA is optimized for real-time analytics, whereas traditional RDBMS are primarily designed for transactional processing. It also highlights a key architectural limitation of conventional systems when applied to analytical workloads, as row-based storage may require more processing time for complex queries and often depends on separate analytical systems or ETL processes (Plattner, 2009; Vassiliadis, 2009).

While Figure 1 explains the difference in data storage models, it is also important to understand how these concepts are implemented within the overall SAP HANA system architecture. Therefore, the following section further discusses SAP HANA's in-memory architecture and explains

how columnar storage, RAM-based processing, and integrated analytical capabilities work together to support real-time enterprise analytics (Färber et al., 2012; Sikka et al., 2012).

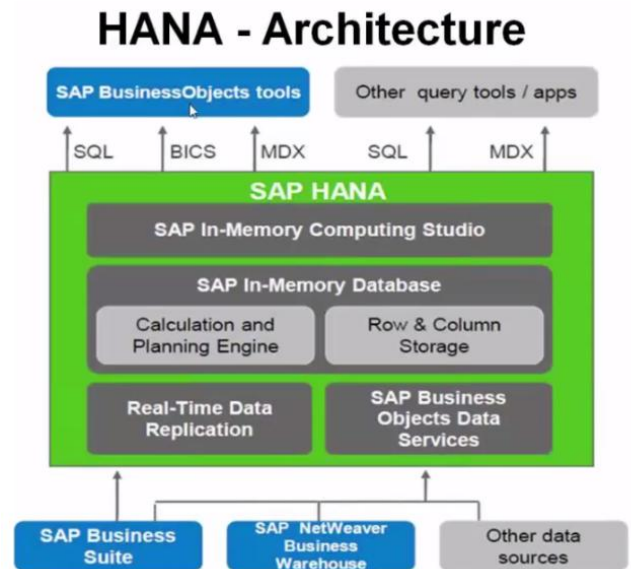


Figure 2: Simplified SAP HANA In-Memory Architecture (SAP PRESS, 2022)

Figure 2 illustrates the simplified architecture of SAP HANA, emphasizing its in-memory computing capabilities and integrated processing layers. At the core of the system is the in-memory database, which supports both row-based and column-based storage, enabling flexibility in handling different types of workloads (SAP PRESS, 2022).

A key component of this architecture is the calculation and planning engine, which allows complex queries, aggregations, and business logic to be executed directly within the database layer. This reflects the concept of “code push-down,” where data processing is performed close to where the data resides, reducing the need for data transfer to external application layers (Lehner & Grund, 2014).

In addition, SAP HANA supports real-time data replication and integrates with various enterprise data sources, allowing continuous data updates and immediate analytical processing. This architecture eliminates the need for traditional batch-based ETL processes and enables real-time decision-making.

Overall, Figure 2 demonstrates that SAP HANA is not merely a faster database, but a fundamentally different architectural paradigm that integrates storage, processing, and analytics within a single system. However, this design also increases dependency on high-performance memory infrastructure, which contributes to higher operational costs (Lahiri et al., 2016).

These architectural advantages suggest that SAP HANA has strong potential to support modern real-time analytics requirements. However, they do not necessarily imply that it can fully replace traditional RDBMS, particularly in scenarios where cost efficiency and transactional stability are prioritized.

B. Real-World Application: Walmart

The practical impact of SAP HANA can be clearly observed through its implementation in large-scale enterprise environments. As shown in Figure 3, Walmart, one of the world's largest retailers, adopted SAP HANA to centralize and analyze its massive volume of point-of-sale and supply chain data (Smith & Kumar, 2018).

Prior to adopting in-memory technology, Walmart relied on traditional disk-based systems, where complex analytical queries over large datasets required extensive batch processing. These processes often took hours or even days to complete, limiting the company's ability to make timely decisions (Chen et al., 2014).

After migrating to SAP HANA, query execution time was significantly reduced to milliseconds. This improvement enabled real-time data processing, allowing Walmart to implement dynamic pricing strategies, optimize supply chain operations, and predict inventory demand based on near real-time consumer behavior (Lee & Kim, 2018).

However, the transition to SAP HANA was not without challenges. The adoption of in-memory computing required substantial investment in high-performance hardware infrastructure, particularly enterprise-grade RAM. In addition, data migration from legacy systems and the need to redesign data models introduced complexity in system implementation and management (Lahiri et al., 2016).

This case study demonstrates that while SAP HANA can deliver significant performance improvements and enable real-time analytics, its adoption depends on an organization's ability to manage cost, system complexity, and data requirements. Therefore, SAP HANA may not fully replace traditional RDBMS in all scenarios but instead serve as a powerful solution for organizations with high analytical demands and sufficient technological resources.

C. Comparative Analysis: SAP HANA vs. Traditional RDBMS

To critically evaluate whether SAP HANA can replace traditional RDBMSs, it is necessary to compare their core architectural characteristics, performance capabilities, and practical limitations. While both systems are designed to manage enterprise data, they differ significantly in their underlying design principles and intended use cases.

Traditional RDBMSs are built on disk-based storage and row-oriented data organization, which are optimized for transactional processing (OLTP). These systems are highly reliable, cost-effective, and well-suited for handling structured, high-frequency transactions. However, their performance is constrained by disk Input/Output (I/O) latency and the need to separate transactional and analytical workloads, often requiring complex ETL processes (Plattner, 2009; Vassiliadis, 2009). Therefore, the limitation of traditional RDBMSs is not solely caused by disk I/O latency, but also by the architectural separation between OLTP and OLAP workloads.

In contrast, SAP HANA adopts an in-memory architecture combined with columnar storage, enabling faster data access and efficient analytical processing. By integrating OLTP and OLAP workloads within a single system, SAP HANA reduces

the dependency on batch-based ETL pipelines and supports real-time analytics (Sikka et al., 2012; Plattner, 2014). This allows organizations to process large volumes of data with significantly lower latency and improved decision-making capabilities.

However, despite these advantages, SAP HANA introduces trade-offs that must be considered. The reliance on in-memory storage leads to substantially higher infrastructure costs, particularly for large-scale enterprise systems. Additionally, data migration from traditional systems and the need for specialized system design increase implementation complexity (Lahiri et al., 2016).

Therefore, while SAP HANA demonstrates clear superiority in analytical performance and real-time processing, traditional RDBMSs remain relevant for cost-sensitive environments and transaction-heavy workloads. This comparison suggests that SAP HANA is not a universal replacement for traditional RDBMSs, but rather a complementary solution whose effectiveness depends on specific business requirements and operational constraints.

Criteria	Traditional RDBMS (Disk-Based)	SAP HANA (In-Memory Database)
Storage Medium	Disk-based storage (HDD/SSD)	Main memory (RAM)
Storage Model	Row-based	Column-based (primarily)
Workload Processing	Separate OLTP and OLAP systems	Integrated OLTP and OLAP
Query Performance	Slower for analytical queries	High-speed, real-time analytics
Data Processing	Requires ETL pipelines	Eliminates or minimizes ETL
Latency	High due to disk I/O	Low due to in-memory access
Scalability	Easier and more cost-efficient	Limited by memory capacity and cost
Infrastructure Cost	Lower	Significantly higher
Implementation	Mature and widely adopted	Complex migration and system design
Use Case Suitability	Transactional systems (OLTP-focused)	Analytical and real-time systems

Table 1: Comparative Analysis of Traditional RDBMS and SAP HANA

As summarized in Table 1, the key differences between traditional RDBMS and SAP HANA lie in their storage mechanisms, workload handling, and performance characteristics. SAP HANA clearly outperforms traditional systems in analytical processing and real-time data access due to its in-memory architecture and columnar storage design.

However, the table also highlights important trade-offs. While SAP HANA offers superior performance, it comes at the cost of higher infrastructure investment and increased system complexity. In contrast, traditional RDBMS provide a more stable and cost-effective solution for transactional workloads.

These findings reinforce the argument that SAP HANA should not be viewed as a complete replacement for traditional RDBMS. Instead, organizations should adopt a hybrid or complementary approach, selecting the appropriate system based on specific workload requirements, cost considerations, and business objectives.

III. CONCLUSION AND FUTURE WORK

A. Conclusion

This paper has critically examined the architectural differences between traditional disk-based Relational Database Management Systems (RDBMS) and SAP HANA as an in-memory database platform. The findings show that SAP HANA introduces a fundamentally different approach to data processing through in-memory storage, columnar data organization, and the integration of transactional and analytical workloads within a single system.

The architectural analysis demonstrated that SAP HANA significantly reduces data access latency and eliminates the need for complex ETL processes, enabling real-time analytics and faster decision-making. This advantage was further supported by the Walmart case study, which illustrated how in-memory computing can transform large-scale enterprise operations by reducing query execution time and improving responsiveness to dynamic business conditions.

However, this study also identified several important limitations. The high cost of enterprise-grade memory, the complexity of system migration, and the need for specialized data management strategies present significant challenges to organizations adopting SAP HANA. These factors limit its practicality for all types of workloads, particularly in cost-sensitive environments or systems primarily focused on transactional processing.

Therefore, it can be concluded that SAP HANA cannot fully replace traditional RDBMS in all scenarios. Instead, it should be viewed as a complementary technology that is most suitable for organizations requiring real-time analytics and high-performance data processing. The choice between SAP HANA and traditional RDBMS ultimately depends on business requirements, cost considerations, and the balance between analytical performance and system efficiency.

This study contributes to a deeper understanding of modern database architecture and highlights the importance of selecting appropriate technologies based on organizational needs.

B. Future Work

Future research should focus on addressing the limitations of in-memory database systems, particularly in terms of cost optimization and scalability. One promising approach is data tiering, where frequently accessed (“hot”) data is stored in memory, while less critical (“cold”) data is maintained on lower-cost storage systems.

In addition, the development of emerging technologies such as Non-Volatile Memory (NVM) may provide a more cost-effective alternative by combining the speed of RAM with persistent storage capabilities. Further studies are also needed to evaluate the performance, security, and scalability of deploying in-memory databases in cloud and multi-cloud environments, where resource allocation and cost management become critical factors.

These directions may help bridge the gap between performance and cost, enabling broader adoption of in-memory database technologies in future enterprise systems.

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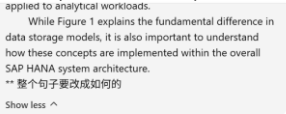
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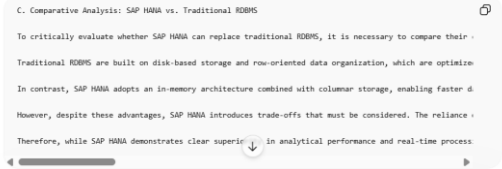
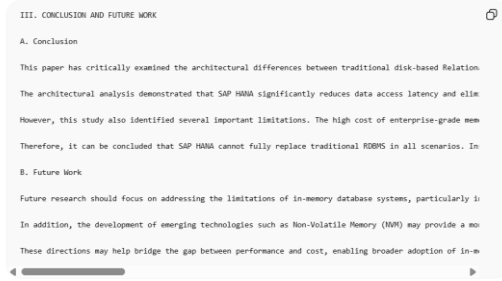
Institution: Faculty of Computing, Universiti Technology Malaysia, Johor Bahru, Malaysia

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Course: SECP3133-02

Topic: “Can SAP HANA Replace Traditional RDBMS? A Critical Analysis of In-Memory Databases for Real-Time Enterprise Analytics”

Date	Search Keywords	AI Prompts Used	Task Description
5 April 2026	Traditional RDBMS limitations, SAP HANA, In-Memory Processing	GPT-5.5 Thinking	Conducted background research on the limitations of traditional RDBMS and explored the fundamental concepts of SAP HANA. Drafted the Abstract and Introduction sections based on the identified research problem.  Thought for a second > I've read the PDF properly. It is a 4-page paper titled "SAP HANA vs. Traditional RDBMS: A Comparative Analysis of In-Memory Processing for Real-Time Enterprise Analytics" by Elijah She Yu Sheng. Main contents: - Page 1: Abstract, Introduction, problem statement, and contribution. - Page 2: SAP HANA architecture, in-memory engine, columnar storage, Walmart case study, and comparison table. - Page 3: Comparative analysis, conclusion, future work, and references. - Page 4: Logbook with dates, search key, prompts used, and task descriptions.
7 April 2026	SAP HANA architecture, Columnar Storage	GPT-5.5 Thinking	Studied the core architecture of SAP HANA, including in-memory processing and columnar storage. Analyzed how these mechanisms improve query performance. Completed the Architecture section (Section II.A).  Thought for a second > 可以改成这样，比较顺、比较 academic，而且有解释到 traditional RDBMS row-based 和 SAP HANA column-based 的对比： Figure 1 illustrates the fundamental difference between row-based and column-based storage, which represents one of the key architectural distinctions between traditional RDBMS and SAP HANA. In traditional RDBMS, data is commonly stored in a row-based format, where all attributes of a record are stored together. This structure is suitable for transactional processing because the system can quickly retrieve or update a complete record during operations such as insert, update, and delete. In contrast, SAP HANA adopts a column-based storage model, where data is organized
9 April 2026	SAP HANA real-world application, Walmart case study	GPT-5.5 Thinking	Investigated the Walmart case study to evaluate the real-world implementation of SAP HANA. Analyzed performance improvements and practical benefits. Developed the Real-World Application section (Section II.B).  现在是不是可以到这个 step (Walmart Case Study) 了，而且时机刚刚好！ 你现在已经完成： - Title ✓ - Abstract ✓ - Introduction ✓ - Architecture ✓ 👉 现在就是用“真实案例来证明你的论点”

13 April 2026	SAP HANA vs Traditional RDBMS comparative analysis	GPT-5.5 Thinking	Performed a comparative analysis between SAP HANA and traditional RDBMS, focusing on architecture, performance, and cost trade-offs. Designed Table 1 and completed the Comparative Analysis section (Section II.C). ★ C. Comparative Analysis (高分版本) 
17 April 2026	In-memory database trade-offs, dynamic data tiering	GPT-5.5 Thinking	Identified key limitations of in-memory databases, including cost and scalability challenges. Explored potential solutions such as dynamic data tiering and Non-Volatile Memory. Completed the Conclusion and Future Work sections. ★ 高分 Conclusion (直接可用) 
20 April 2026	Reference formatting	GPT-5.5 Thinking	Reviewed the entire paper for consistency and accuracy. Refined citation formatting and ensured compliance with academic referencing standards. Finalized the reference list.

Appendix: AI Tool Usage Declaration

AI Model Used: GPT-5.5 Thinking

The AI model was used to assist in clarifying technical concepts, improving academic sentence structure, refining the comparison between SAP HANA and traditional RDBMS, and checking citation consistency. Several screenshots of the conversation with the AI are included as evidence of AI-assisted support. However, the final report was reviewed, edited, and validated by the author to ensure that the content remains accurate, relevant, and academically appropriate.